

# John Doe

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## EDUCATION

### Private Liberal Arts College

*Bachelor of Science in Computer Science & Mathematics*

**GPA:** 3.67 / 4.00

**Relevant Coursework:** Algorithms, Advanced Algorithms, Data Structures, Randomized Algorithms for Data Science, Big Data, Discrete Mathematics, Abstract Algebra

United States

*Aug 2023 – May 2027*

## EXPERIENCE

### Machine Learning Engineering Intern

*Public Research University*

Sep 2025 – Present

*Remote*

- Designed and implemented a Python data pipeline to ingest, validate, and structure 200+ survey responses for automated analysis.
- Built a FastAPI backend exposing REST endpoints to orchestrate preprocessing, embedding, and topic-labeling workflows.
- Reduced end-to-end processing latency from approximately 5 minutes to under 1 minute using batching and vector-based indexing.
- Improved system reliability by adding structured logging, schema validation, and robust failure handling.

### Undergraduate Research Assistant

*Private STEM College*

May 2025 – Aug 2025

*United States*

- Implemented and benchmarked Python algorithms for NP-complete problems, including structured partitioning and sumset decision variants.
- Developed a testing framework generating 10,000+ randomized instances to evaluate correctness and average-case performance.
- Achieved a 20% runtime improvement using preprocessing, pruning heuristics, and optimized branching strategies.
- Analyzed worst-case versus average-case behavior across input distributions to identify regimes where exact algorithms outperform heuristics.

### Computer Science Teaching Assistant

*Private STEM College*

Aug 2024 – Present

*United States*

- Supported 200+ students by debugging and optimizing Python programs, diagnosing performance, memory, and logic errors.
- Created reusable debugging guides and test scaffolds adopted across multiple course sections.

## PROJECTS

### Chatbot | *Python, FastAPI, React, WebSockets, LangChain, ChromaDB*

Oct 2024 – May 2025

- Built a full-stack, real-time chat application with a FastAPI backend and WebSocket-based token streaming.
- Designed an asynchronous, multi-user backend with session management and concurrent request handling.
- Implemented a document ingestion and vector search pipeline enabling semantic retrieval across 200+ documents.
- Reduced ingestion and retrieval latency from approximately 5 minutes to under 1 minute by batching embeddings and optimizing vector indices.
- Hardened the system for production-style use by adding structured logging, input validation, and recovery mechanisms.

### Food Analyzer | *Python, TensorFlow, NumPy, scikit-learn*

Aug 2025 – Sep 2025

- Built an image classification system for urban scene understanding, covering preprocessing, training, and evaluation.
- Fine-tuned MobileNetV2 on 2,000+ images across 12 classes using data augmentation to mitigate overfitting.
- Achieved 87% test accuracy and evaluated performance using confusion matrices and per-class precision, recall, and F1 metrics.

## TECHNICAL SKILLS

**Languages:** Python, C++, C, Java, JavaScript, TypeScript, SQL, HTML, CSS

**Backend / Systems:** FastAPI, REST APIs, WebSockets, PostgreSQL, Linux, Bash, Git

**Data / ML:** Pandas, NumPy, scikit-learn, TensorFlow, ChromaDB, LangChain, Matplotlib